

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 903/2
Bridge Chainage – 1010+420

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – MJB

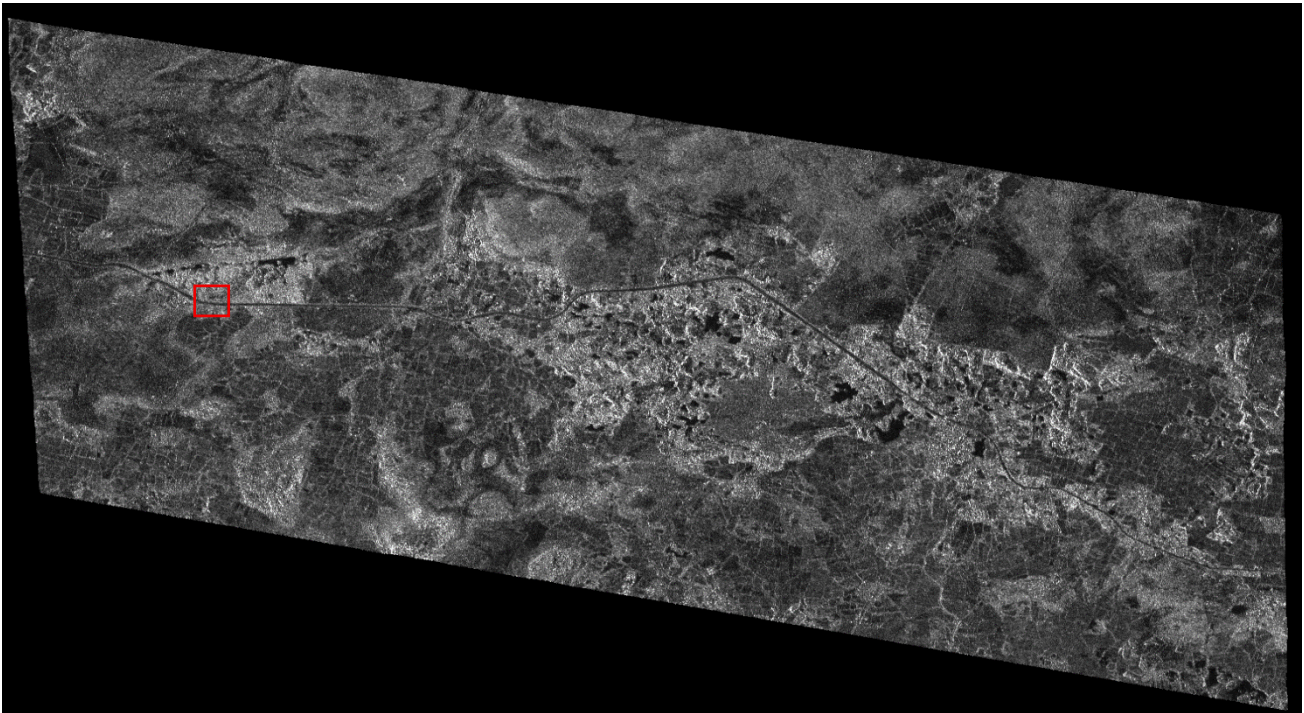
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

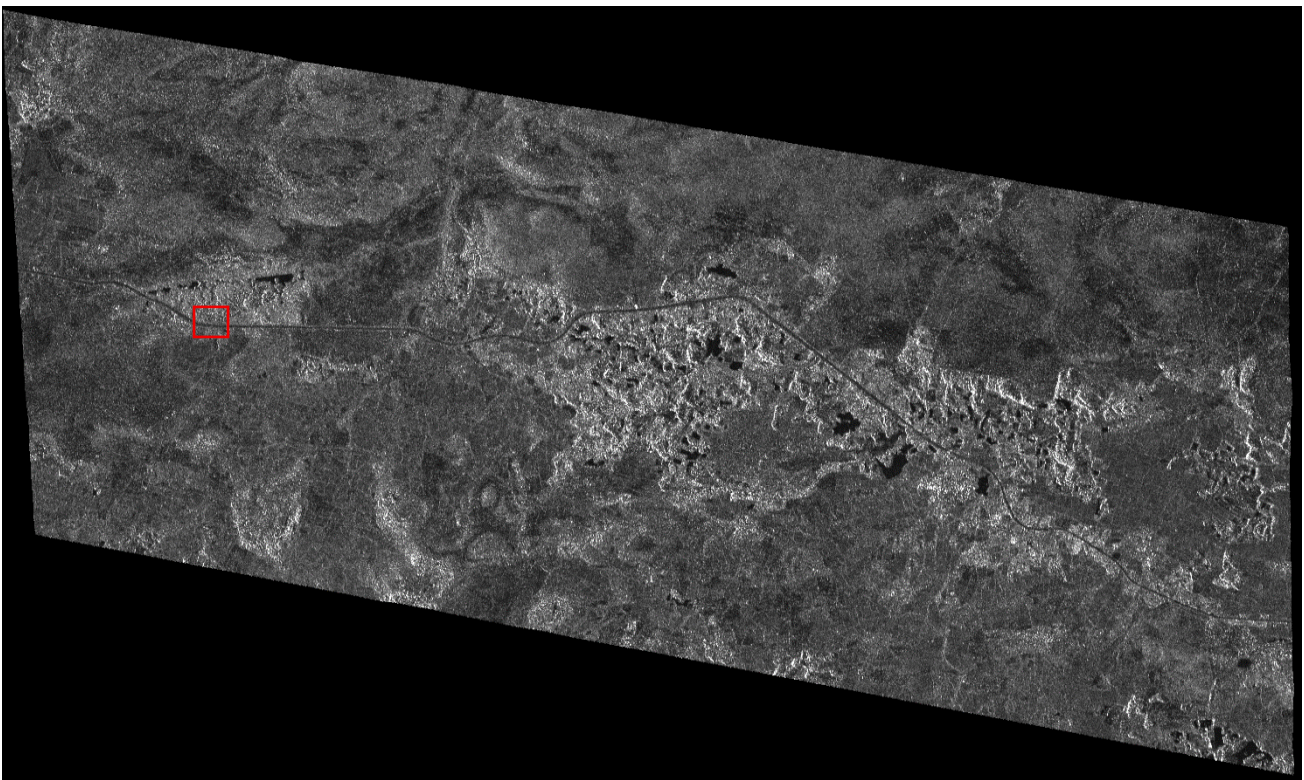
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB – CH-1010+420

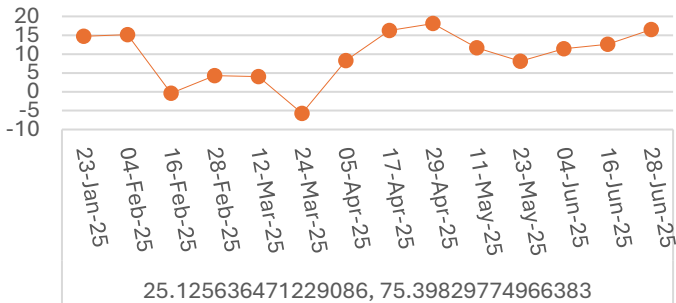
Length – 19.82M



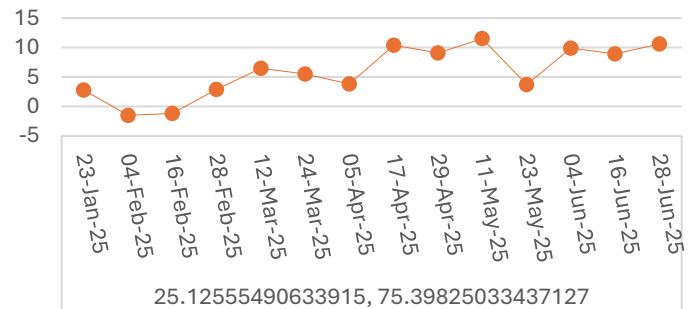
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
P1		Pier 1	
P2		Pier 2	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

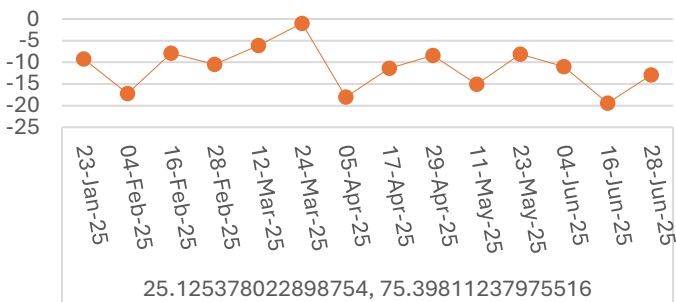
Displacement @ SR A1 LHS VV



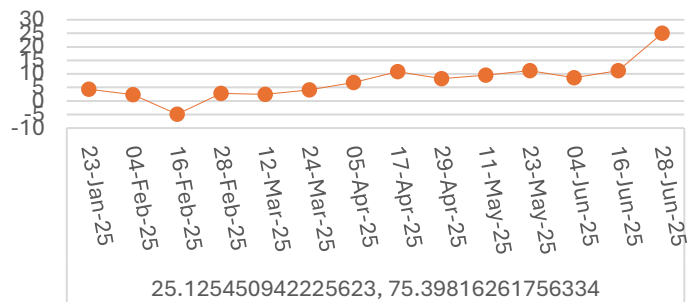
Displacement @ MR A1 LHS VV



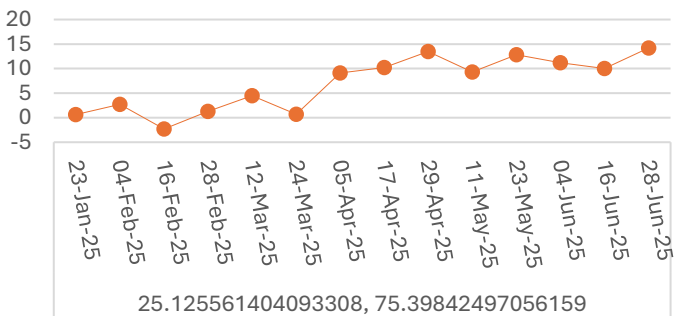
Displacement @ SR A1 RHS VV



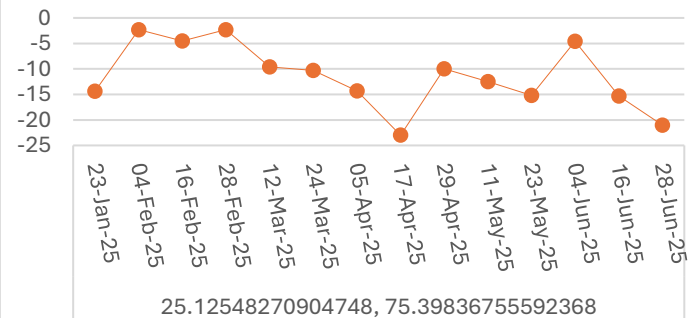
Displacement @ MR A1 RHS VV



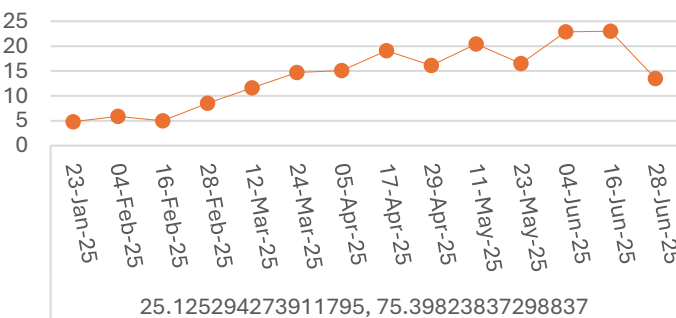
Displacement @ SR A2 LHS VV



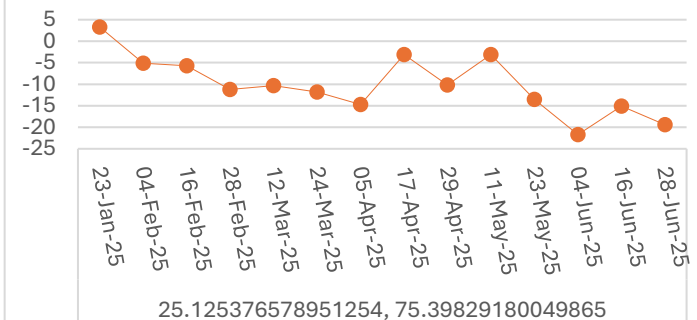
Displacement @ MR A2 LHS VV



Displacement @ SR A2 RHS VV

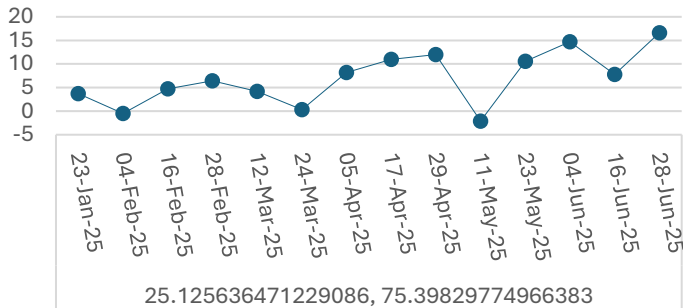


Displacement @ MR A2 RHS VV

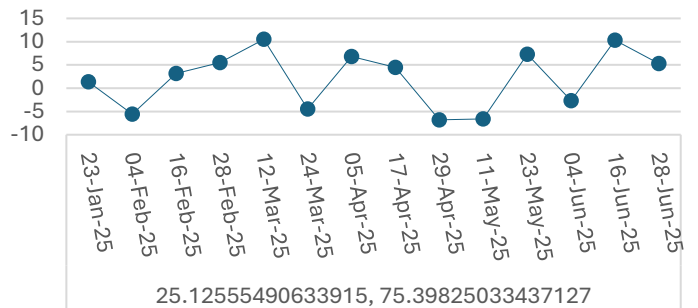


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

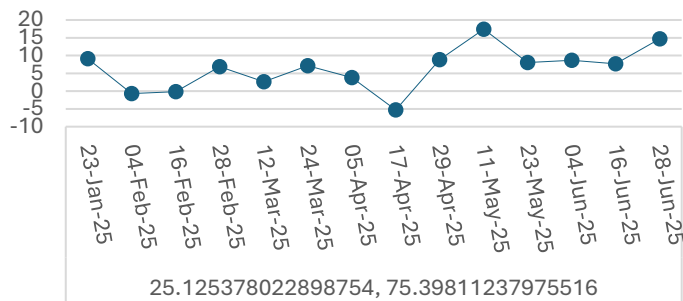
Displacement @ SR A1 LHS VH



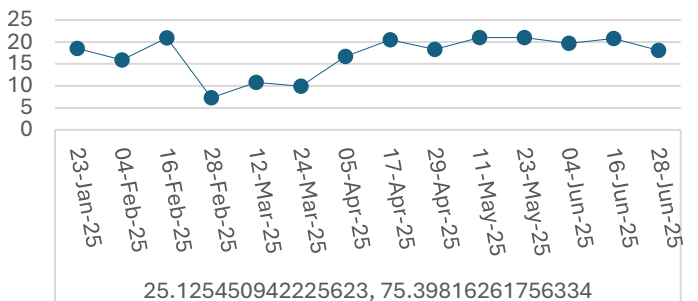
Displacement @ MR A1 LHS VH



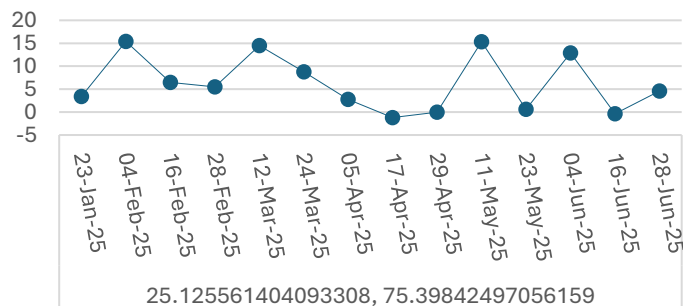
Displacement @ SR A1 RHS VH



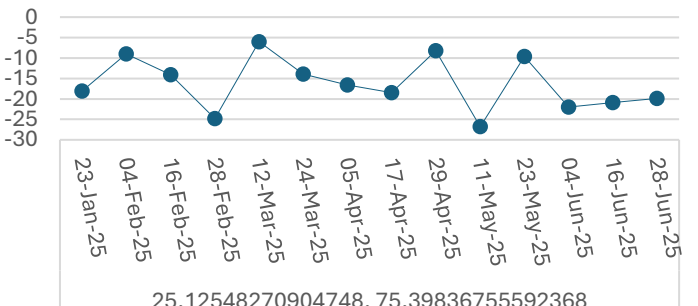
Displacement @ MR A1 RHS VH



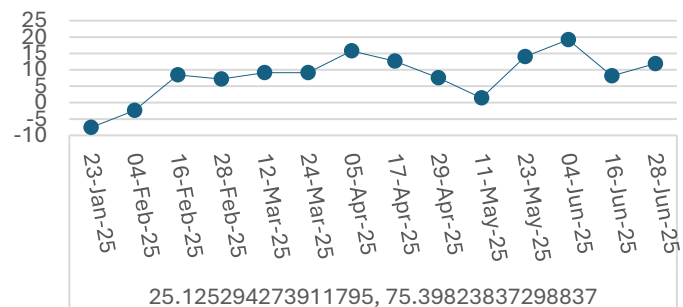
Displacement @ SR A2 LHS VH



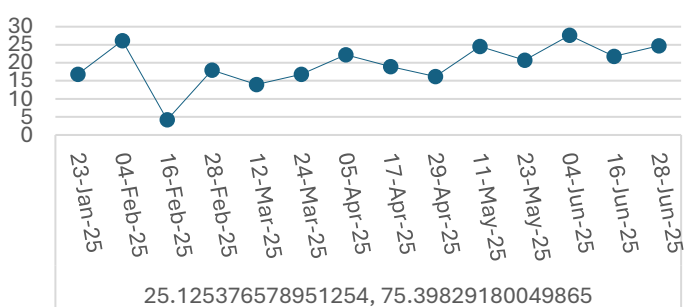
Displacement @ MR A2 LHS VH



Displacement @ SR A2 RHS VH



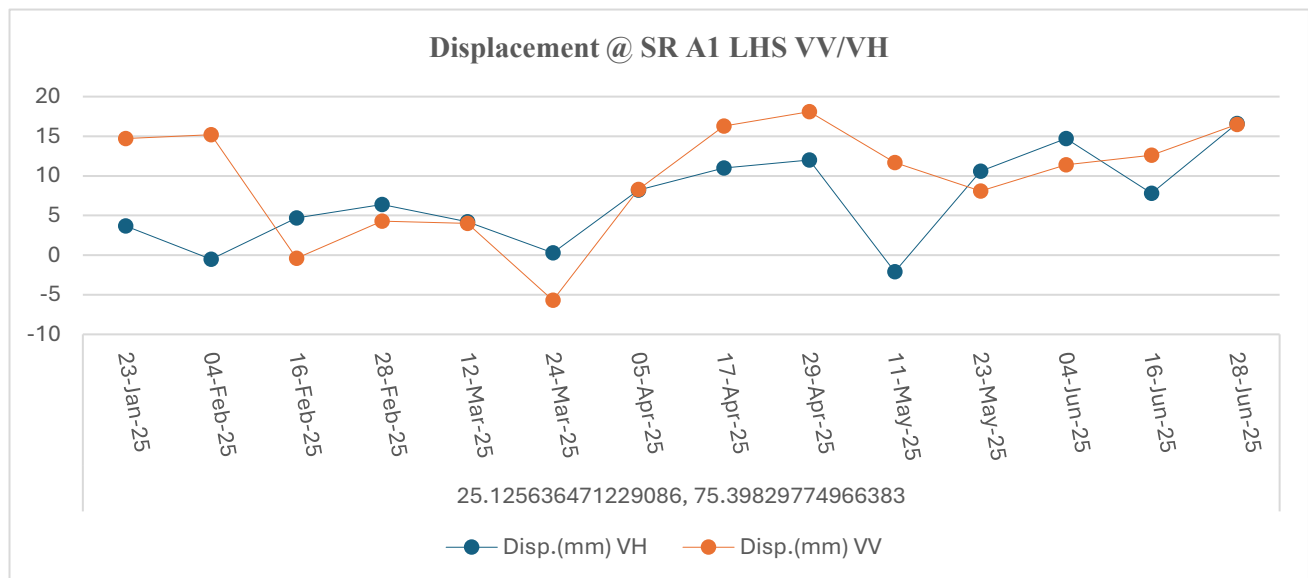
Displacement @ MR A2 RHS VH



3.3 SAR Based Analysis at Service Road Abutment 1 LHS:



SR LHS A1 (25.125636471229086, 75.39829774966383)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows moderate to high variability with an overall positive trend over the monitoring period.
- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate oscillatory behaviour, varying roughly between -3 mm and +15 mm.

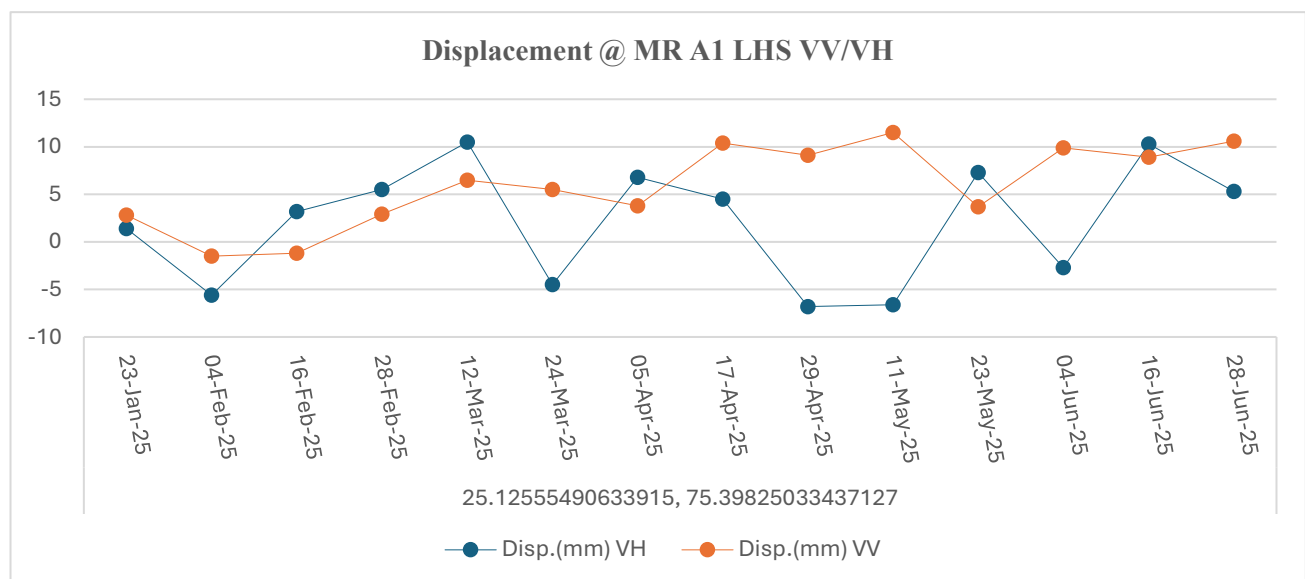
Critical Events & Anomalies:

- **March 2025:** VV reaches a distinct negative minimum while VH also decreases to 0–1 mm, marking a short-term coupled downward adjustment affecting both displacement components.
- **April–May 2025:** VV peaks near +18 mm, while VH stabilizes around +11–12 mm, highlighting a strong vertical uplift dominance during this interval.
- **June 2025:** VH and VV representing the largest combined vertical uplift and lateral displacement of the monitoring period, indicating continued deformation activity at this abutment location.

3.4 SAR Based Analysis at Main Road Abutment 1 LHS:



MR LHS A1 (25.12555490633915, 75.39825033437127)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows high variability with frequent oscillations, ranging roughly between -7 mm and $+10$ mm.
- **Vertical displacement (VV)** Vertical displacement exhibits moderate variability with a generally positive trend over the monitoring period (Jan–Jun 2025).

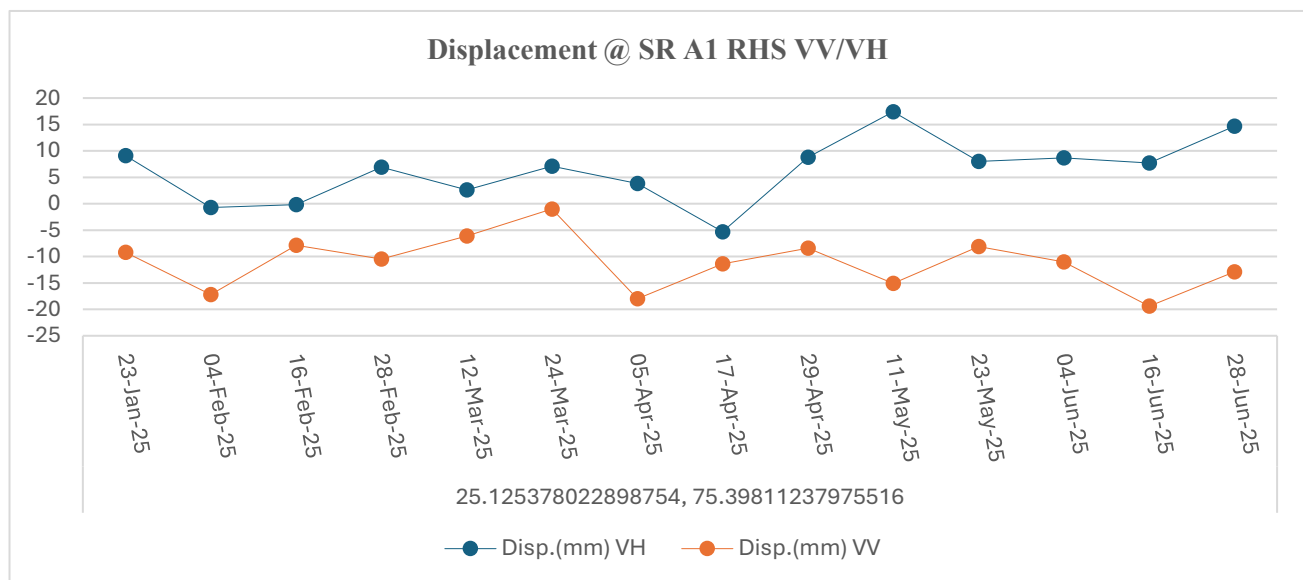
Critical Events & Anomalies:

- **February 2025:** VH drops to around -6 mm, while VV also becomes slightly negative (-2 mm), marking a short-term coupled settlement event affecting both horizontal and vertical components.
- **April–May 2025:** VH reaches a pronounced negative minimum (-7 mm), while VV peaks near $+11$ mm, representing a strongly asymmetric deformation phase.
- **June 2025:** Both VH and VV end at moderate positive values (VH $+5$ mm, VV $+10$ mm), suggesting partial stabilization but ongoing deformation activity at this location.

3.5 SAR Based Analysis at Service Road Abutment 1 RHS:



SR RHS A1 (25.125378022898754, 75.39811237975516)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows consistently negative values throughout the monitoring period (Jan–Jun 2025), indicating persistent downward settlement at the SR A1 RHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits moderate to high variability, ranging roughly between –5 mm and +18 mm.

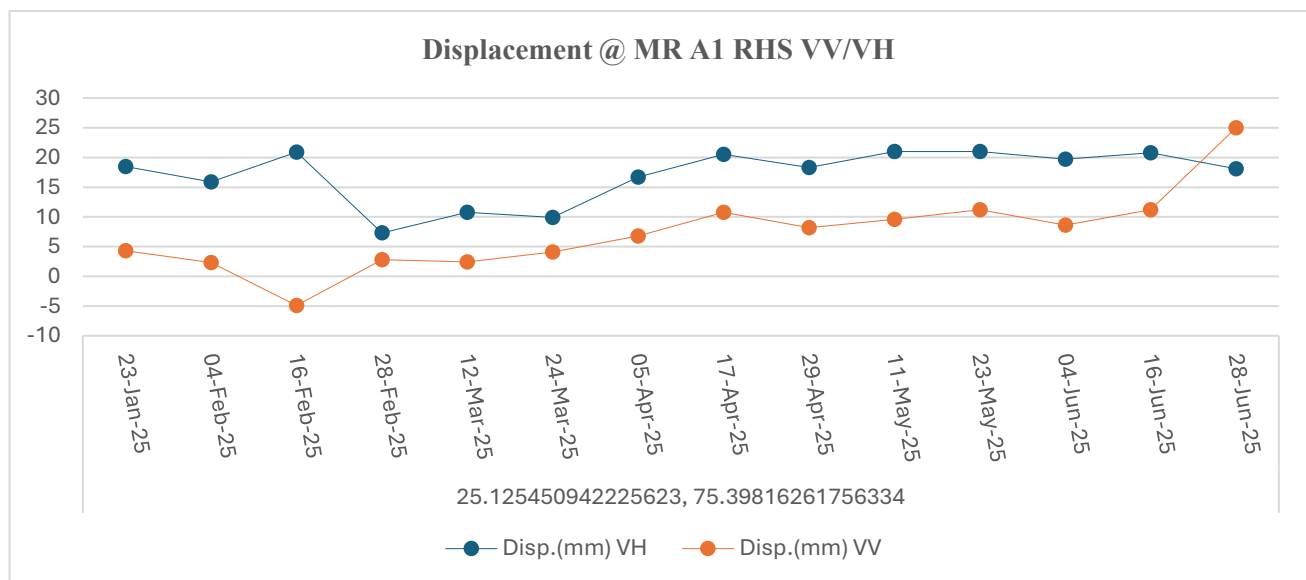
Critical Events & Anomalies:

- **February 2025:** VV drops sharply to around –18 mm, while VH remains near 0 mm, indicating a strong vertical settlement event without corresponding lateral movement.
- **April 2025:** VH rebounds sharply to +8 mm, while VV improves slightly but remains strongly negative, indicating lateral recovery without vertical stabilization.
- **June 2025:** VH ends at a high positive value (+15 mm), while VV remains negative (-14 mm), confirming ongoing deformation activity with no vertical recovery at this abutment location.

3.6 SAR Based Analysis at Service road Abutment 1 RHS:



MR RHS A1 (25.125450942225623, 75.39816261756334)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows a clear transition from early-period settlement to strong uplift toward the end of the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits consistently positive values with moderate variability, ranging roughly between +8 mm and +21 mm.

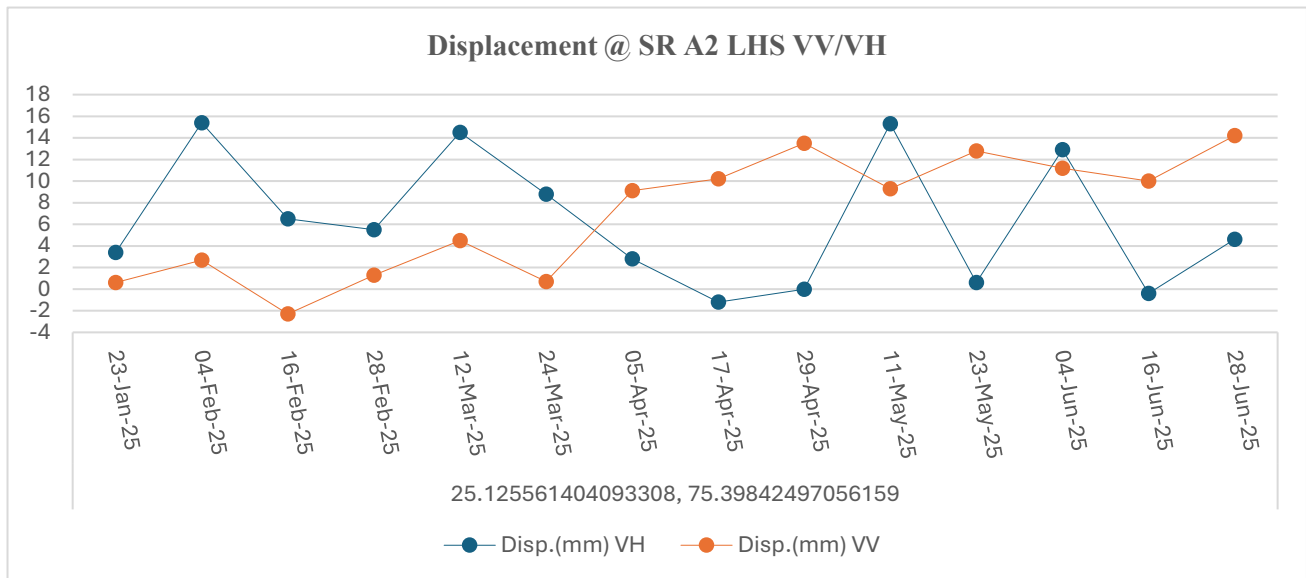
Critical Events & Anomalies:

- **February 2025:** VH remains elevated (+15–18 mm), while VV decreases to –5 mm, indicating a temporary vertical settlement phase occurring under continued lateral displacement.
- **May 2025:** VH remains consistently high (+19–21 mm), while VV fluctuates between +7 mm and +10 mm, reflecting a stable lateral displacement regime with moderate vertical uplift.
- **June 2025:** VH shows minimal variation (+19 mm), while VV increases again to +14 mm, suggesting continued vertical accumulation under steady lateral movement.

3.7 SAR Based Analysis at Service Road Abutment 2 LHS:



SR LHS A2(25.125561404093308, 75.39842497056159)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement exhibits high variability with frequent oscillations, ranging roughly between -2 mm and $+15$ mm.
- **Vertical Displacement (VV)** Vertical displacement shows a predominantly positive trend throughout the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the SR A2 LHS location.

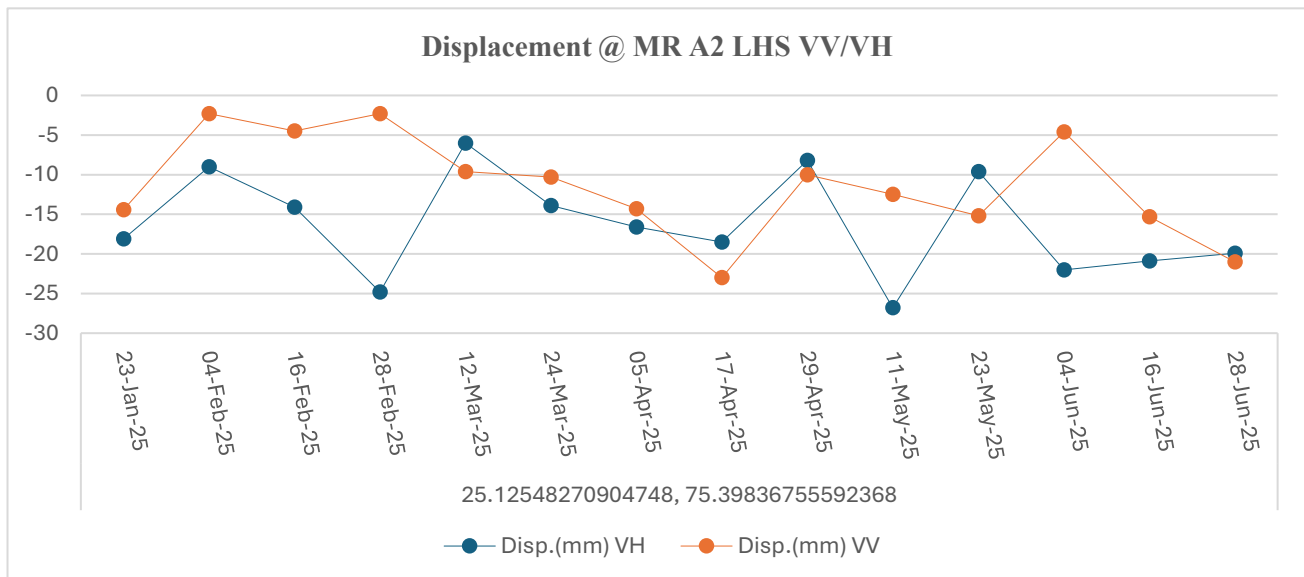
Critical Events & Anomalies:

- **February 2025:** VH reaches a local positive peak ($+15$ mm), while VV remains low but positive ($+3$ mm), indicating a strong lateral displacement event with limited vertical participation
- **May 2025:** VH drops again to $+2$ mm, while VV remains elevated ($+13$ mm), suggesting temporary lateral relaxation without vertical loss.
- **June 2025:** VH recovers to $+4$ – 5 mm, while VV peaks again near $+14$ mm, marking renewed uplift toward the end of the monitoring period.

3.8 SAR Based Analysis Main Road Abutment 2 LHS:



MR LHS A2(25.12548270904748, 75.39836755592368)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement is predominantly negative throughout the monitoring period (Jan–Jun 2025), indicating persistent vertical settlement at the MR A2 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits high variability with strong negative dominance, ranging roughly between –8 mm and –27 mm.

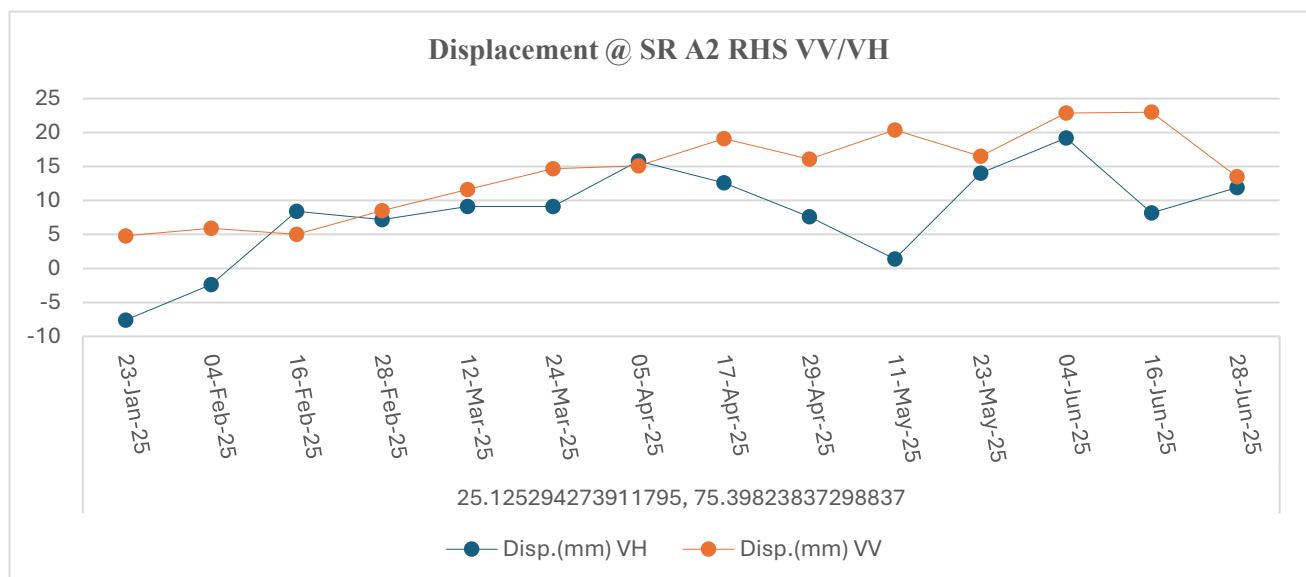
Critical Events & Anomalies:

- **March 2025:** VH recovers to –6 mm, while VV remains negative (–9 mm), suggesting temporary lateral relaxation without vertical stabilization.
- **June 2025:** Both VH and VV converge near –20 to –22 mm, indicating persistent combined settlement with no recovery trend toward the end of the monitoring period.

3.9 SAR Based Analysis at Service Road Abutment 2 RHS:



SR RHS A2 (25.125294273911795, 75.3982383729883)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows a consistently positive and increasing trend throughout the monitoring period (Jan–Jun 2025), indicating dominant vertical uplift at the SR A2 RHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits moderate to high variability, ranging roughly between -7 mm and $+19$ mm.

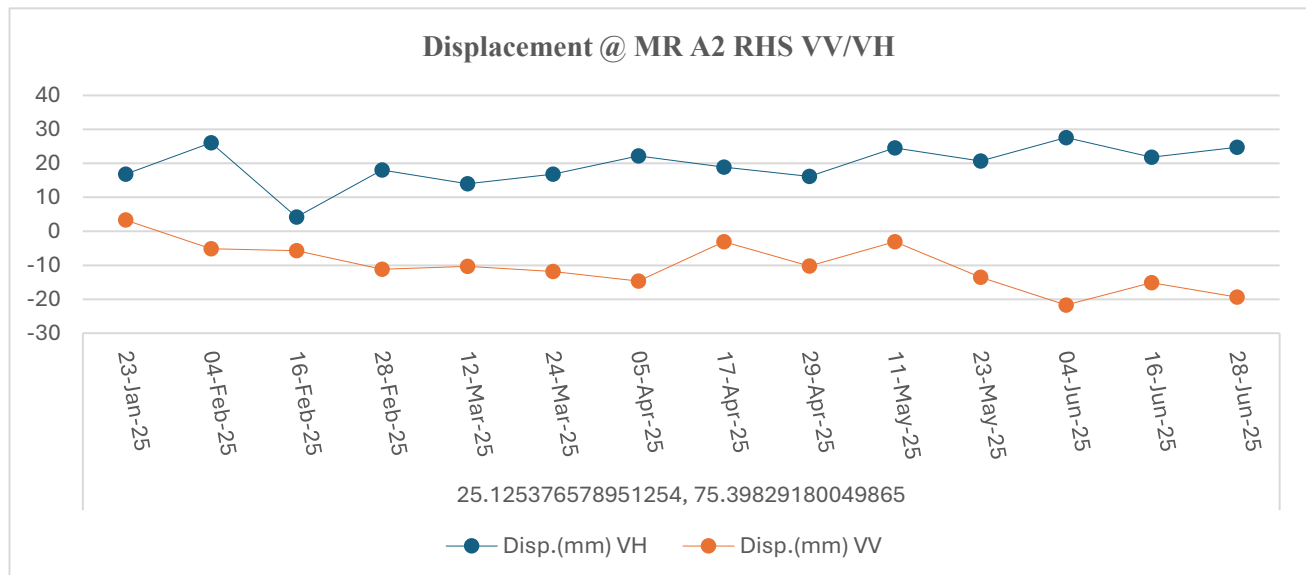
Critical Events & Anomalies:

- **February 2025:** VH crosses into positive values ($+8$ mm), while VV increases to $+7$ mm, marking the onset of coupled vertical uplift and lateral displacement.
- **June 2025:** VH decreases slightly to $+8$ – 12 mm, while VV remains high ($+13$ – 14 mm), indicating post-peak adjustment under sustained uplift conditions

3.10 SAR Based Analysis at Main Road Abutment 2 RHS:



MR RHS A2 (25.125376578951254, 75.39829180049865)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement is consistently negative throughout the monitoring period (Jan–Jun 2025), indicating persistent vertical settlement at the MR A2 RHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits consistently positive values with moderate variability, ranging roughly between +5 mm and +28 mm.

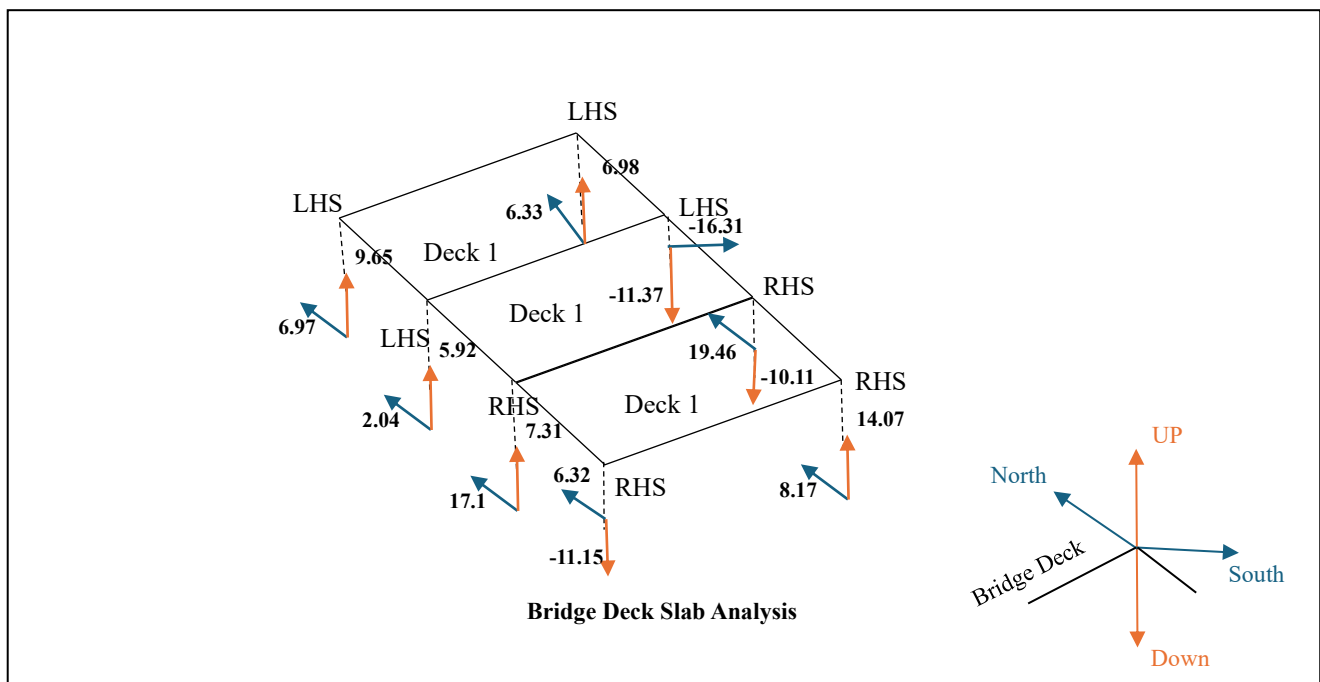
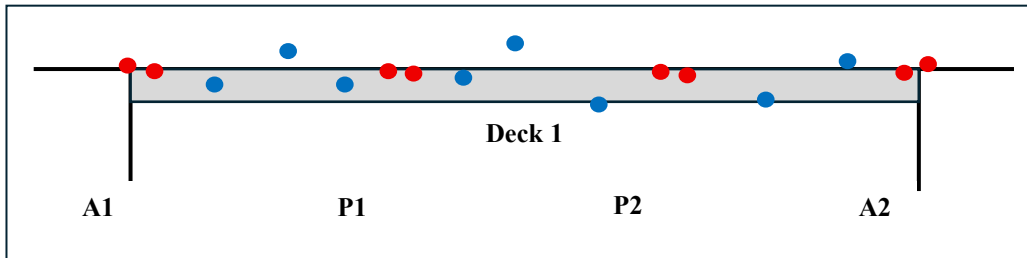
Critical Events & Anomalies:

- **Feb 2025:** VH rises sharply to +25 mm, while VV drops to –5 mm, indicating a strong lateral displacement event occurring alongside moderate vertical settlement.
- **May 2025:** VV drops sharply to –20 to –25 mm, while VH remains high (+20–28 mm), representing the most severe vertical settlement phase coupled with dominant lateral movement.
- **June 2025:** VH remains consistently elevated (+22–25 mm), while VV stays strongly negative (–15 to –20 mm), indicating persistent deformation with no vertical stabilization toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- Vertical displacement across the bridge deck slab exhibits pronounced differential behaviour, ranging from +9.65 mm uplift at the LHS upper deck point to -11.37 mm downward displacement at the central RHS upper deck point. Additional significant settlement is observed at the lower RHS deck location (-11.15 mm), while moderate uplift is recorded at the lower RHS outer point (+14.07 mm).

Horizontal Displacement Behaviour:

- Horizontal displacement magnitudes vary substantially, ranging from +19.46 mm at the central RHS deck point to -16.31 mm at the LHS upper-right deck edge, confirming strongly opposing lateral movement directions.

CONCLUSION:

- The combined vertical and horizontal displacement patterns indicate a highly non-symmetric deformation response of the bridge deck slab. The LHS is characterised by consistent uplift with relatively smaller horizontal movement, whereas the RHS exhibits significant lateral displacement combined with alternating uplift and settlement zones. The coexistence of uplift on one side and settlement on the opposite side further reinforces the likelihood of rotational deformation rather than uniform translation.